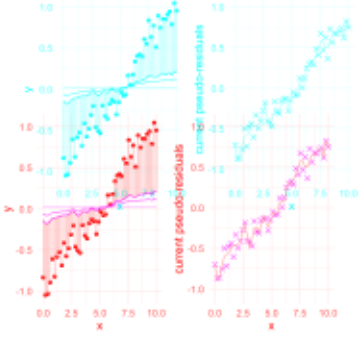
Boosting

Gradient Boosting: Illustration

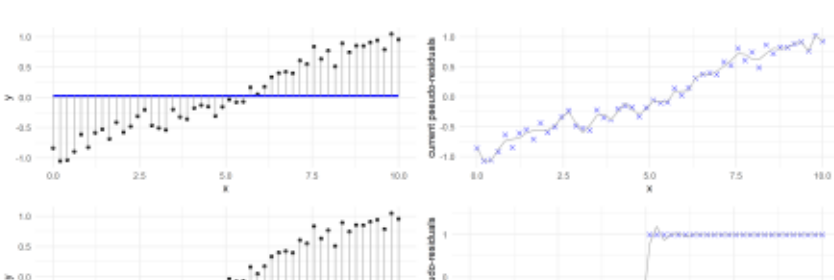


Learning goals

- See simple visualizations of boosting in regression
- Understand impact of different losses and base learners

GAM WITH $L2$ VS $L1$ LOSS

Top: $L2$ loss, bottom: $L1$ loss



Iteration 1

Shape of PRs affects gradual model fit: $L1$ only sees resids' sign, BLs are not affected size of values as in $L2$ and hence lead to more moderate changes.

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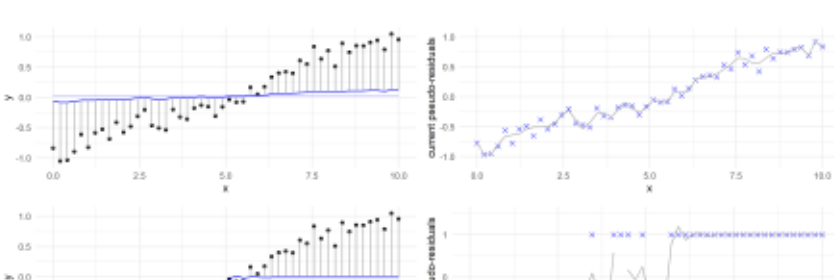

Iteration 2

Shape of PRs affects gradual model fit: $L1$ only sees resids' sign, BLs are not affected size of values as in $L2$ and hence lead to more moderate changes.

GAM WITH $L2$ VS $L1$ LOSS

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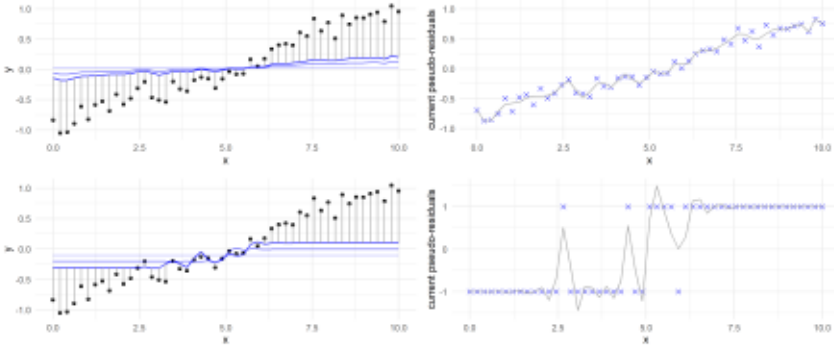
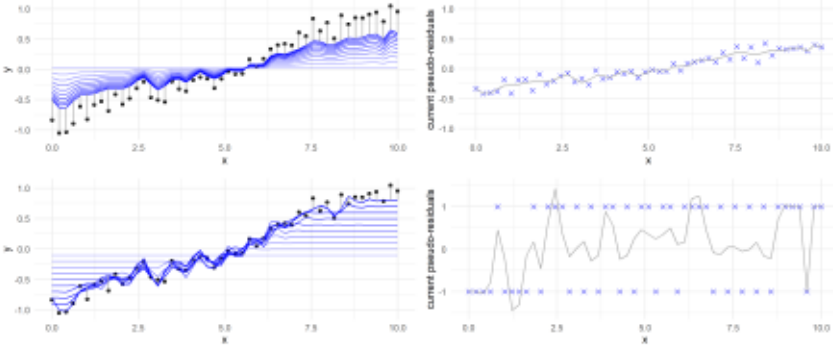


Iteration 3

Shape of PRs affects gradual model fit: $L1$ only sees resids' sign, BLs are not affected size of values as in $L2$ and hence lead to more moderate changes.

GAM WITH $L2$ VS $L1$ LOSS

Top: $L2$ loss, bottom: $L1$ loss

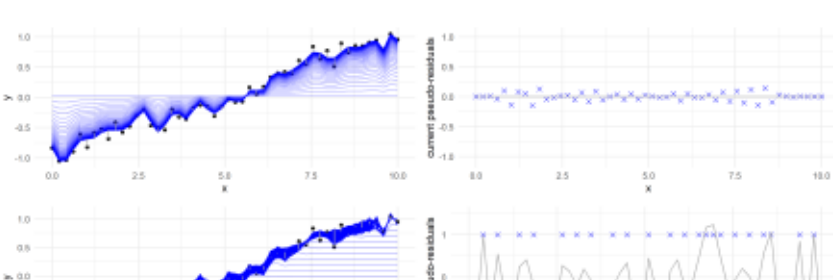


Iteration 10

Shape of PRs affects gradual model fit: $L1$ only sees resids' sign, BLs are not affected size of values as in $L2$ and hence lead to more moderate changes.

GAM WITH $L2$ VS $L1$ LOSS

Top: $L2$ loss, bottom: $L1$ loss

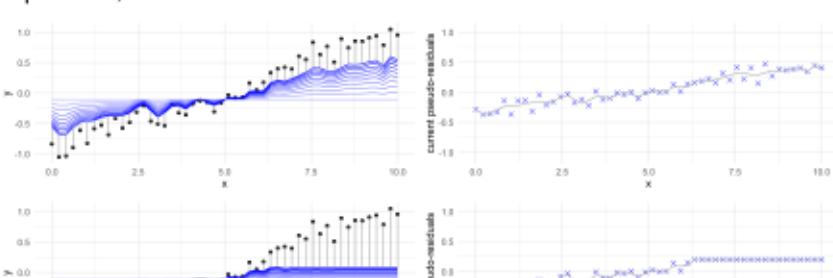


Iteration 100

Shape of PRs affects gradual model fit: $L1$ only sees resids' sign, BLs are not affected size of values as in $L2$ and hence lead to more moderate changes.

GAM WITH HUBER LOSS

Top: $\delta = 2$, bottom: $\delta = 0.2$.

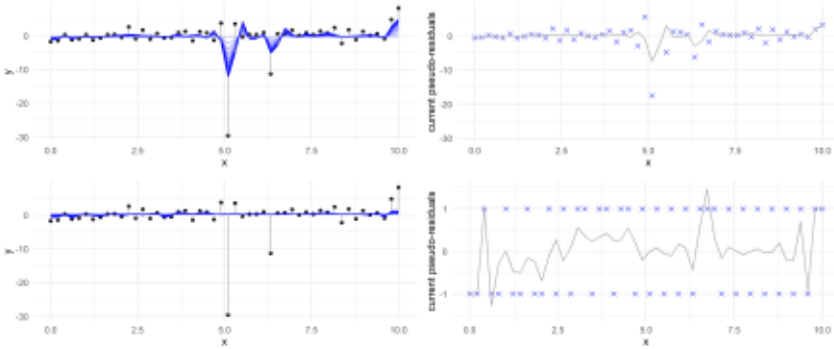


Iteration 10

- For small δ, PRs are often bounded, resulting in $L1$-like behavior, while the upper plot more closely resembles $L2$ loss.

GAM WITH OUTLIERS

- Instead of Gaussian noise, let's use t-distrib, that leads to outliers in y. Top: $L2$, bottom: $L1$.



Iteration 10

$L2$ loss is affected by outliers rather strongly, whereas $L1$ solely considers residuals' sign and not their magnitude, resulting in a more robust model.

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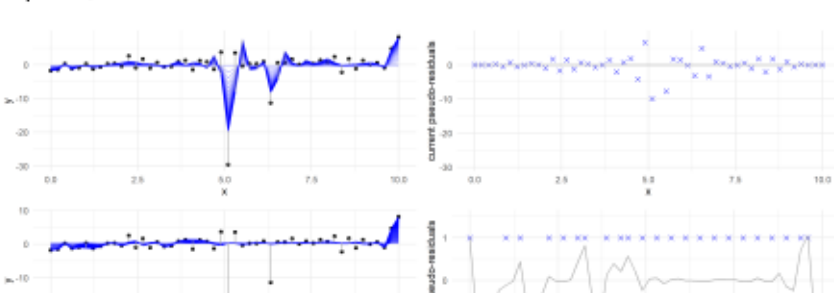
Instead of Gaussian noise, let's use t-distrib, that leads to outliers in y. Top: $L2$, bottom: $L1$.



Iteration 100

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